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S.Y. B.Tech
(TERM- III)
CAE I- WINTER 2025 (2024 Pattern)
Data Structures and Algorithms (25BTPCL302)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) *All questions are compulsory.*
ii) *Neat diagrams must be drawn wherever necessary.*
iii) *Figures to the right indicate full marks.*
iv) *Use of scientific calculator is not allowed*
v) *Assume suitable data, if necessary*

Q.No	Question	Marks	CO	BL
Q1.(a)	Define time complexity and space complexity. Analyze the time and space complexity of a linear search algorithm.	[5]	CO1	L2
b)	Apply shell sort to sort the following list of integers. What is the time complexity of shell sort? (Write the output of each iteration clearly) li = [23, 12, 1, 8, 34, 54, 2, 3]	[5]	CO2	L3
	OR			
Q 2.(a)	Apply radix sort to sort the following list of integers. What is the time complexity of radix sort? (Write the output of each iteration clearly) li = [170, 45, 75, 90, 802, 24, 2, 66]	[5]	CO1	L3
(b)	Compare linear search vs binary search. Explain the concept of data modification in an array.	[5]	CO2	L2
Q 3.(a)	Explain the concepts of a linear array. Write a complete C function int get Element By Index (int arr[], int size, int index) that safely retrieves an element by its index, handling out-of-bounds errors. Demonstrate its use in main.	[5]	CO1	L3
(b)	Compare Singly Linked List vs Circular Doubly Linked List.	[5]	CO2	L2
	OR			
Q. 4 (a)	Describe the concept of an Array as an Abstract Data Type (ADT). List its basic operations and explain any two.	[5]	CO1	L2
(b)	Explain insertion (at beginning and in empty list) and deletion (at second position and at last node) operation in a doubly linked list.	[5]	CO2	L3